classical limit. A particle with low mass, moving through an abrupt potential, behaves like a quantum mechanical system.

What about electrons? Are they massive enough to behave classically? The answer depends on the interplay between the potential and the mass. For example, if you have two capacitor plates separated by a centimeter, with a smooth electric field between them, then the electron will move across the gap like a nice, coherent, almost classical particle. On the other hand, the potential associated with the nucleus of an atom always has a sharp feature in it. If an electron wave packet hits this potential, it will scatter all over the place.

Before leaving this topic, I'd like to mention *minimum-uncertainty* wave packets. These are wave packets where $\Delta x \Delta p$ is *equal* to $\hbar/2$ (as opposed to being greater). In other words, in these cases, $\Delta x \Delta p$ is as small as quantum mechanics allows. These wave packets have the form of a Gaussian curve, and they're often called *Gaussian* wave packets. Over time, they spread out and flatten. Such wave packets are not that common, but they do exist. A bowling ball at rest is a good approximation. In Lecture 10, we'll see that the ground state of a harmonic oscillator is a Gaussian wave packet.

9.8 PATH INTEGRALS

Classical Hamiltonian mechanics focuses on the step-by-step incremental changes in the state of a system. But there is another way to formulate mechanics—the Principle of Least Action—in which the focus is on entire histories. For a particle, this means looking at the full trajectory of the particle from some initial time to some final time. The content of the two approaches is the same, but the emphasis is different. Hamiltonian mechanics zeros in on some instant and tells you how the system changes between that instant and the next. The least action principle steps back and takes a global look. One can imagine nature sampling all possible trajectories and picking the one that minimizes the action between a pair of fixed initial and final points.[7]